

MBS-004 - Trigger-Retrieval-Tunnel Architecture

A Structural Runtime Model for MET Brain Structure

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Repository: <https://github.com/sizhet/MET-Brain-Structure-MBS>

Introduction

The MET Brain Structure (MBS) project proposes that many apparently different cognitive phenomena may emerge from a common structural runtime architecture.

Dreaming.

Memory recall.

Danger response.

Navigation.

Skill acquisition.

Imagination.

Learning.

Planning.

Cognitive convergence.

Although these phenomena appear different at the behavioral level, they may share common structural foundations.

The purpose of this document is not to describe biological implementation details.

Those remain the domain of neuroscience.

Instead, this document proposes a structural and computational abstraction capable of organizing a broad range of observations within a unified framework.

This abstraction is called the Trigger-Retrieval-Tunnel Architecture (TRTA).

The Core Hypothesis

The central hypothesis is simple:

Biological cognition may be organized around the interaction of five major layers:

1. Trigger Layer
2. Retrieval Layer
3. Graph and Tunnel Knowledge Layer
4. Embodied Runtime Layer
5. In-Situ Evolution Layer

Together these layers form a continuously operating structural runtime system.

Why a Layered Model?

Many traditional computational models implicitly assume:

Input

→ Processing

→ Output

While useful in engineering, this view struggles to explain:

- Instant recognition
- Reflexive reactions
- Dream generation
- Intuition
- Skill execution
- Context-dependent retrieval

In many situations, cognition appears to operate through activation rather than construction.

The brain often does not appear to build behavior from scratch.

Instead, it appears to activate, retrieve, combine, and adapt existing structures.

The Trigger-Retrieval-Tunnel Architecture is designed to capture this observation.

Layer 1

Trigger Layer

The Trigger Layer is responsible for detecting significant situations.

Examples include:

- Danger
- Opportunity
- Hunger
- Social signals
- Novelty
- Goals
- Emotional cues
- Environmental changes

The Trigger Layer acts as a high-priority attention and activation system.
Its primary purpose is not detailed reasoning.

Its purpose is rapid relevance detection.

A trigger answers a simple question:

What deserves attention now?

From an evolutionary perspective, this may be one of the oldest components of biological cognition.

Trigger Convergence

One important implication follows.

Different organisms often encounter similar survival pressures.

As a result, they may evolve similar trigger categories.

Examples include:

- Predator detection
- Food identification
- Mate recognition
- Offspring protection
- Territory awareness

This may explain why cognitive convergence is often observed across individuals and species.

Similar triggers can produce similar retrieval behaviors.

Layer 2

Retrieval Layer

Once a trigger activates, cognition faces a second problem:

Finding relevant knowledge quickly.

This is the role of the Retrieval Layer.

The Retrieval Layer may include:

- Classification structures
- Similarity recognition systems
- Memory indexing mechanisms
- Pattern matching processes
- Attention allocation systems

Its purpose is not to generate behavior.

Its purpose is to identify candidate structures for activation.

From a Structural Intelligence perspective, this layer resembles:

- Classification Trees
- Differential Trees
- Similarity Search Systems
- Retrieval Graphs

The exact biological implementation remains unknown.
The structural function, however, appears indispensable.

Retrieval as a Survival Mechanism

Fast retrieval often matters more than perfect retrieval.

A prey animal does not need the optimal response.

It needs a sufficient response before being caught.

This observation aligns naturally with the Minimal Evolution Threshold perspective.

Evolution often favors:

Fast enough.

Good enough.

Reliable enough.

Rather than:

Perfect.

Layer 3

Graph and Tunnel Knowledge Layer

The Retrieval Layer identifies candidates.

The Knowledge Layer contains the structures themselves.

MBS proposes that much biological knowledge may be organized as interconnected graph and tunnel systems.

Examples include:

- State relationships
- Action sequences
- Behavioral routines
- Spatial navigation paths
- Social interaction patterns
- Sensorimotor loops

Calling Graph Interpretation

One representation uses Calling Graphs.

State

→ Operation

→ State

Examples:

Threat Detected

→ Escape Action

→ Safe State

Hunger

- Search Behavior
- Food Acquisition

Function Tunnel Interpretation

Another representation uses Function Tunnels.

A tunnel represents a coherent functional pathway.

For example:

Threat Detection

→ Orientation

→ Running

→ Shelter Seeking

→ Recovery

This entire sequence forms a functional tunnel.

Many tunnels may coexist.

Many tunnels may overlap.

Many tunnels may share common segments.

Together they form a Function Tunnel Network.

Why Graphs and Tunnels?

Graphs capture relationships.

Tunnels capture functional continuity.

Both appear repeatedly in biological behavior.

Together they provide a powerful structural language for representing cognition.

Layer 4

Embodied Runtime Layer

The Embodied Runtime Layer is where activated structures become experiences, simulations, or actions.

This layer may support:

- Dreaming
- Imagination
- Prediction
- Planning
- Motor execution
- Memory replay

The runtime layer transforms structural potential into active cognition.

Scene Construction

One of the most important functions of this layer is scene construction.

Scenes may include:

- Objects
- Locations
- Actions
- Goals
- Emotional context
- Expected outcomes

Dreaming provides one of the clearest examples of this capability.

A dream often appears as a dynamically evolving embodied scene rather than a symbolic reasoning process.

Runtime Activation

A simplified flow might be:

Trigger

→ Retrieval

→ Tunnel Activation

→ Scene Construction

→ Runtime Execution

This pattern appears repeatedly across many cognitive activities.

Layer 5

In-Situ Evolution Layer

Biological cognition cannot remain static.

The environment changes continuously.

The organism changes continuously.

Knowledge must therefore evolve continuously.

The In-Situ Evolution Layer supports:

- Learning
- Adaptation
- Reinforcement
- Forgetting
- Structural refinement
- Trigger optimization

This layer operates while the system remains active.

Evolution is not separated from runtime.

Evolution occurs inside runtime.

Dreaming as Structural Maintenance

From the MBS perspective, dreaming may contribute to:

- Tunnel reinforcement

- Trigger refinement
- Retrieval adjustment
- Graph restructuring

Whether or not this interpretation proves fully correct, it provides a compelling explanation for why dreaming might remain evolutionarily valuable.

Unified Runtime Flow

The entire architecture can be summarized as:

Environment

↓

Trigger Detection

↓

Retrieval

↓

Graph/Tunnel Activation

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Embodied Runtime

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Action or Simulation

↓

Learning and Adaptation

↓

Updated Structure

↓

Future Triggering

This cycle never truly stops.

It continuously evolves throughout the lifetime of the organism.

Relationship to Neuroscience

The Trigger-Retrieval-Tunnel Architecture should not be interpreted as a biological claim about specific neurons, circuits, or brain regions.

Instead, it is a structural abstraction.

Its purpose is to provide a conceptual framework capable of organizing diverse observations from neuroscience, cognitive science, psychology, ethology, and artificial intelligence.

The model remains intentionally agnostic regarding biological implementation.

Different neural mechanisms may realize similar structural functions.

Future neuroscience research may confirm, modify, or replace aspects of this

framework.

The value of the architecture lies not in claiming certainty, but in providing a coherent structure for exploration.

Conclusion

The Trigger-Retrieval-Tunnel Architecture proposes that biological cognition may emerge from the interaction of five continuously operating layers:

Trigger

Retrieval

Graph and Tunnel Knowledge

Embodied Runtime

In-Situ Evolution

Together these layers form a structural runtime system capable of supporting dreaming, danger response, memory, learning, imagination, planning, and adaptation.

Rather than viewing cognition as a sequence of computations performed by a central processor, this perspective views cognition as the continuous activation and evolution of interconnected structural pathways.

From the perspective of MET Brain Structure, intelligence may not primarily be about computation.

It may be about the organization, activation, and evolution of structure.